Instance Configuration Template

Generated Files

Instance file	$\|S\|$	$\|P\|$	$\|B\|$	$\|E\|$	$\|T\|$	$\|I\|$	Seed
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$|S|$: students; $|P|$: placements; $|B|$: companies; $|E|$: educational centres; $|T|$: professors; $|I|$: dependency groups; Seed: seed used to generate the instance.

Generator Arguments

Argument	Description	Value
`result_path`	Output folder where the generated files are written	–
`start_size`	First number of students considered	–
`end_size`	Last number of students considered	–
`number_instances`	Number of instances generated for each student size	–
`surplus`	Extra placements percentage with respect to students	–
`slack_ratio`	Extra capacity margin for company and professor quotas	–
`placement_dep_prob`	Percentage used to create placement dependency groups	–
`number_competences`	Number of base competences per profile	–
`most_important_competences`	Number of primary competences required by a placement	–
`seed`	Base seed used to derive the seed of each generated instance	–

Internal Parameters

Parameter	Description	Value
student_size_step	Fixed step between generated student sizes	–
number_profiles	Number of academic profile blocks	–
p_excellent_student	Probability of generating an excellent student	–
p_notable_student	Probability of generating a notable student	–
car_probability	Probability that a student has access to a car	–
student_distance_weight_range	Range used for the student distance weight	–
n_centers_min	Minimum number of educational centres	–
n_centers_max	Maximum number of educational centres	–
prof_capacity_min	Minimum professor supervision capacity	–
prof_capacity_max	Maximum professor supervision capacity	–
ORS_AVAILABLE	Whether ORS routing support is available in the script	–
use_ors	Whether ORS is actually used for this generation	–
ors_base_url	ORS endpoint used when routing is enabled	–
synthetic_duration_mu	Mean of the synthetic duration distribution	–
synthetic_duration_sigma	Standard deviation of the synthetic duration distribution	–

Randomness

Field	Description	Value
Base seed	Initial seed provided to the generator	–
Seed policy	Rule used to derive instance seeds from the base seed	–
Python random seeded	Whether the Python random generator is seeded per instance	–
NumPy random seeded	Whether the NumPy random generator is seeded per instance	–